

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Power input, soft-latch, supplies



File: power.kicad_sch

3-phase power stage and current sensing



File: bridge.kicad_sch

STM32G431 controller



File: mcu.kicad_sch

Connectors and I/O conditioning



File: io.kicad_sch

DESIGN SUMMARY

- Battery: 48 V nominal (13S Li-ion, 54.6 V full). Abs. max 60 V (TVS SMCJ60CA, 100 V FETs/drivers/bucks).
- Battery current limit 35 A (firmware, from FOC power estimate). Phase current limit 60 A (firmware), hard trip 80 A.
- Power stage: 6 x IPT015N10N5 (100 V, 1.5 mOhm, TOLL) driven by 3 x LM5109B half-bridge drivers @ 20 kHz, 4.7 Ohm gate resistors.
- Current sensing: in-line 0.5 mOhm 4-terminal (Kelvin) phase shunts + INA240A1 (G=20, mid-rail ref) -> +/-165 A full scale.
- Control: STM32G431CB @170 MHz, hall-sensor FOC with angle interpolation; hall/R/L auto-detect over UART.
- Supplies: LM5164 48->12 V (gate drive), LM5164 48->5 V (halls/throttle), AP2112 5->3.3 V (logic).
- Soft power: momentary button to GND wakes the bucks via Q1, MCU latches with PWR_HOLD; long press = off.
- I/O: hall + motor NTC (JST-PH6), throttle (PH3), FWD/REV switch (PH2), brake (PH2), power button (PH2), UART 115200 8N1 3.3 V (PH4: 3V3, TX, RX, GND), SWD (1x5 header).

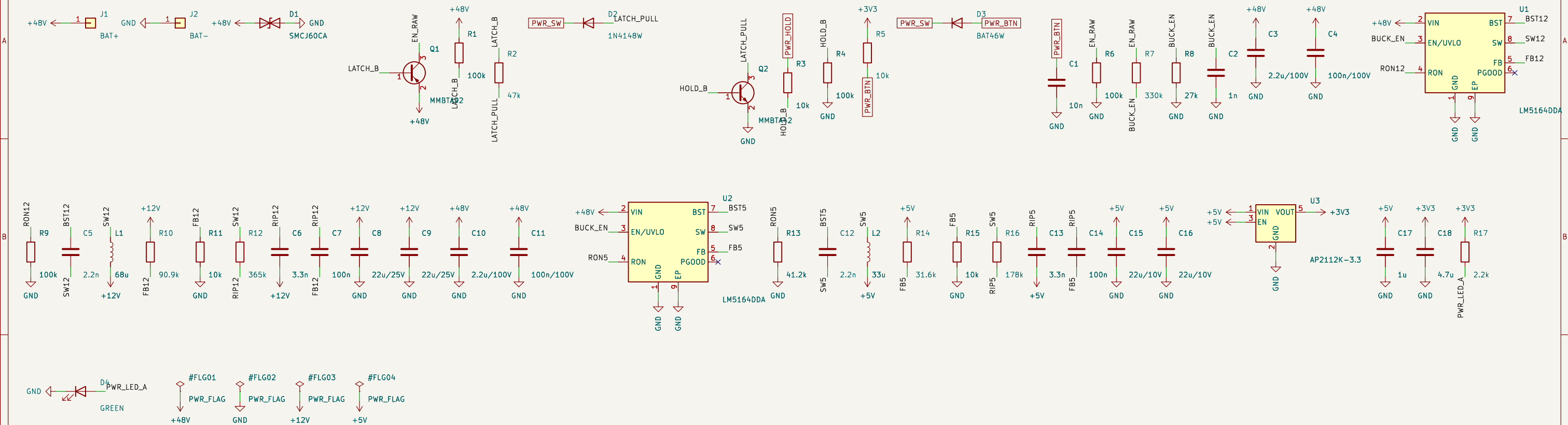
LAYOUT RULES: 2 oz copper min (4 layer preferred), bus/phase pours >= 10 mm wide, FET tabs on thermal-via arrays to an aluminium heat spreader, Kelvin-route INA240 inputs from the shunt pads, DC-link ceramics at each half bridge.

Top level
STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3
48V 35A hub motor controller
Sheet: /
File: bldc48.kicad_sch

Title: 48V 35A BLDC hub motor controller

Size: A3	Date: 2026-10-06	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/5

Power input, soft-latch, supplies

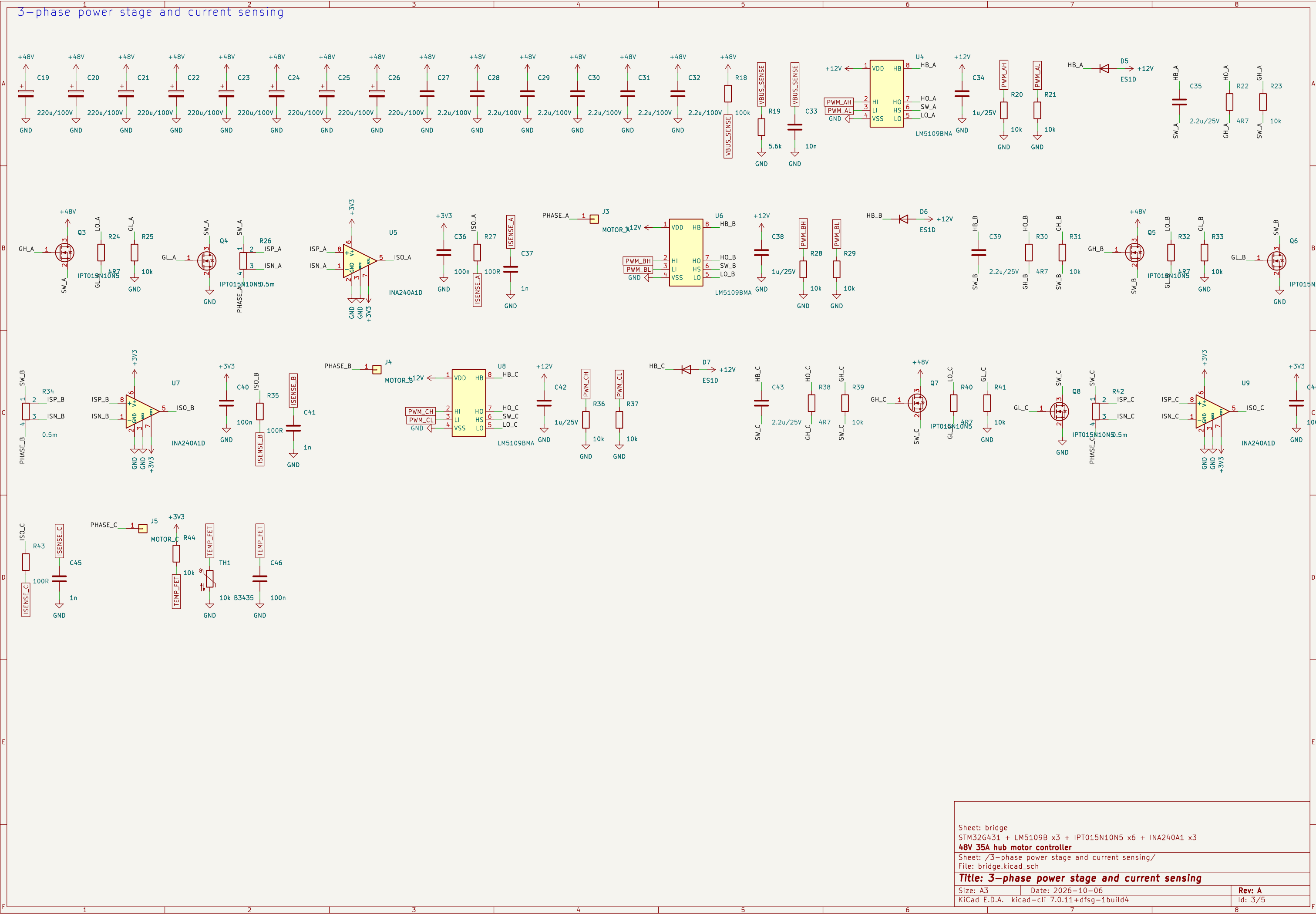


Sheet: power
STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3
48V 35A hub motor controller
Sheet: /Power input, soft-latch, supplies/
File: power.kicad_sch

Title: Power input, soft-latch, supplies

Size: A3	Date: 2026-10-06	Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/5

3-phase power stage and current sensing



Sheet: bridge
STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3
48V 35A hub motor controller
Sheet: /3-phase power stage and current sensing/
File: bridge.kicad_sch

Title: 3-phase power stage and current sensing

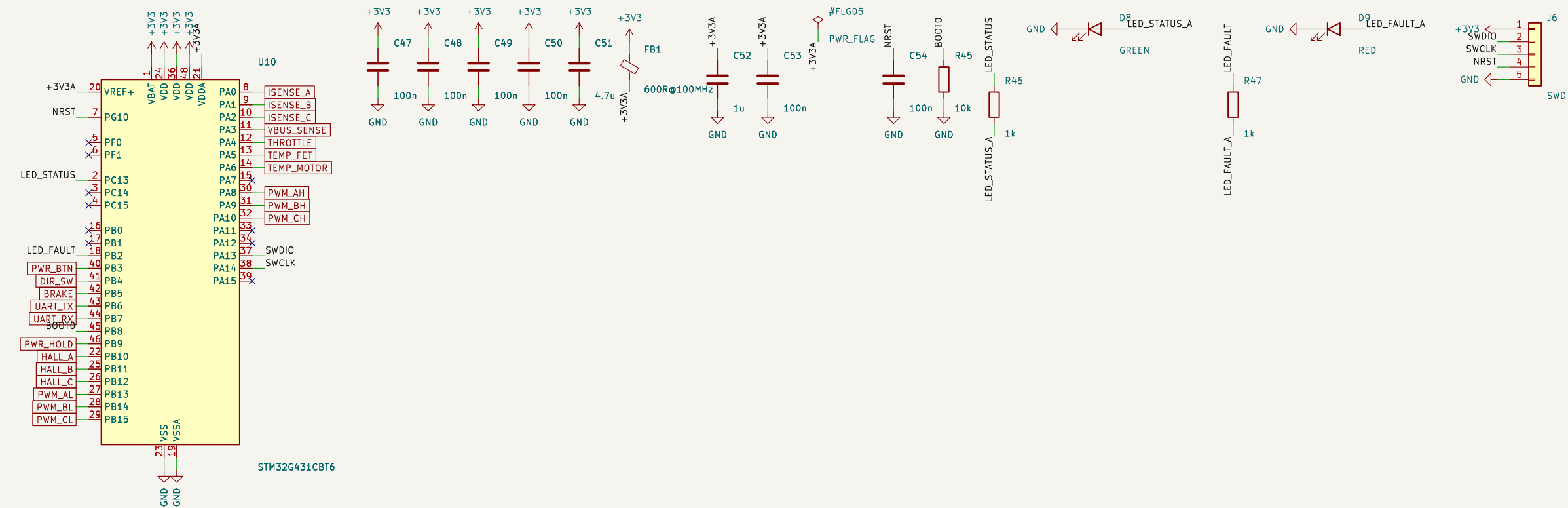
Size: A3 Date: 2026-10-06

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: A

Id: 3/5

STM32G431¹ controller



Sheet: mcu
 STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3
48V 35A hub motor controller
 Sheet: /STM32G431 controller/
 File: mcu.kicad_sch
Title: STM32G431 controller

Size: A3	Date: 2026-10-06	Rev: A
KiCad E.D.A.	kicad-clf 7.0.11+dfsg-1build4	Id: 4/5

Connectors and I/O conditioning

